

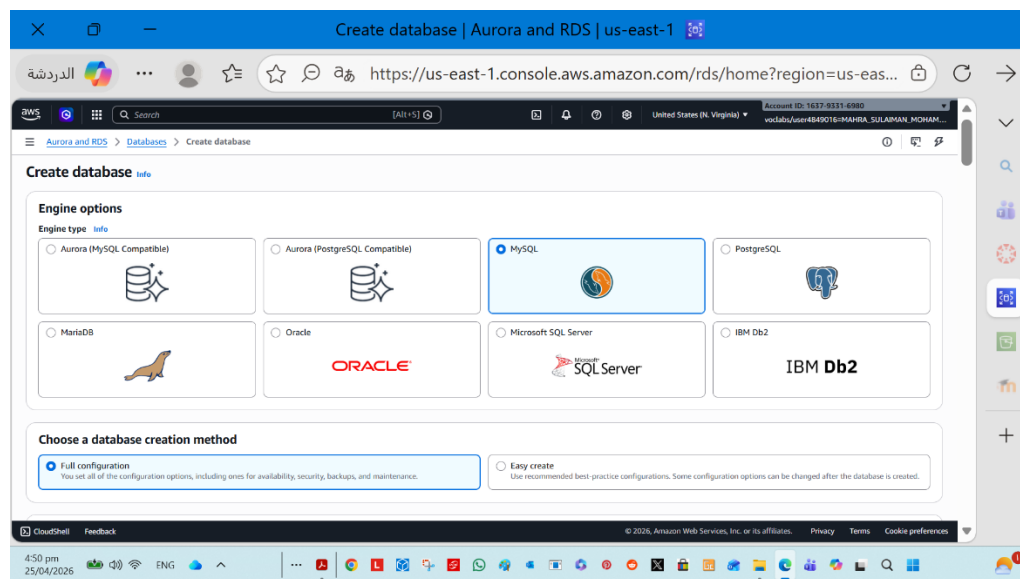
Lab: Build MySQL DB Server using RDS and Interact with DB Using MySQL Workbench

Lab Overview

In this lab, you will create an Amazon Relational Database Service (RDS) MySQL database instance, configure security settings, and verify connectivity using MySQL workbench.

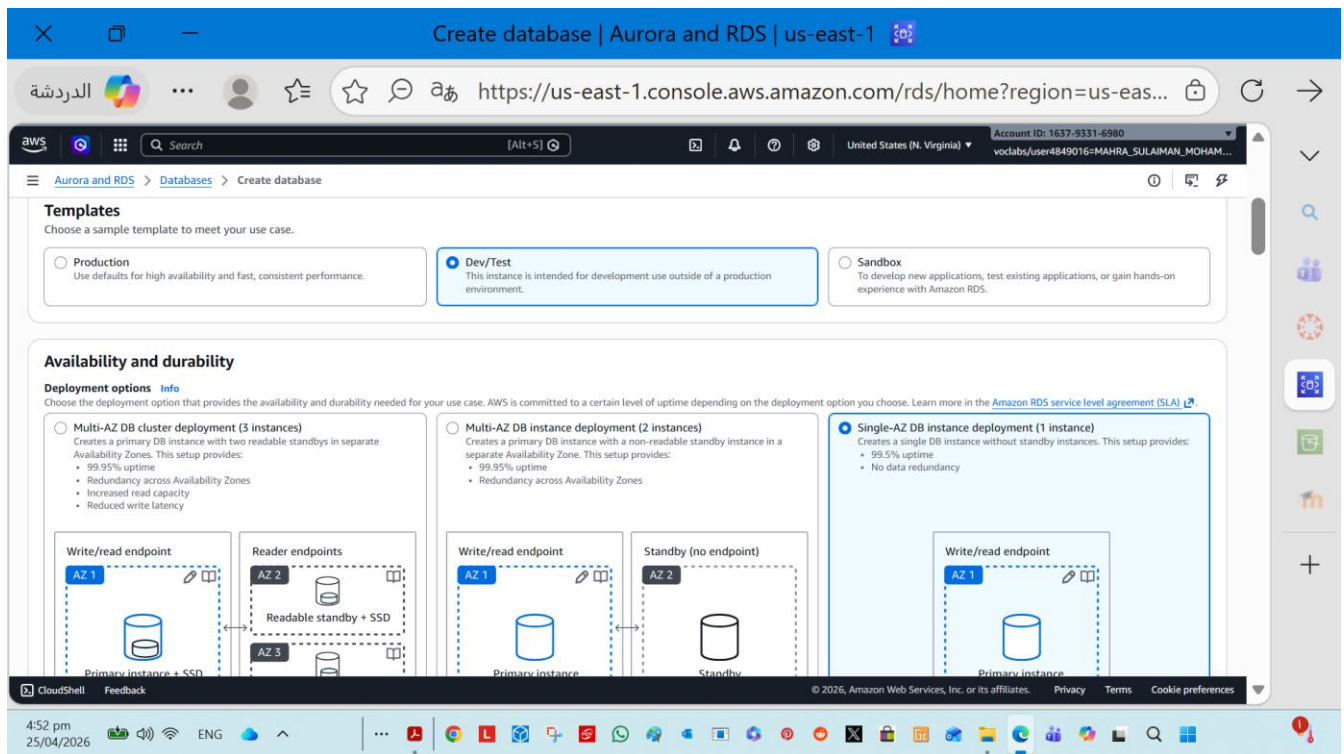
Lab Objectives

- Create an RDS MySQL database instance
 - Configure security group rules for database access
 - Connect to the database using a MySQL client
 - Verify database functionality
 - Clean up resources
- Prerequisites**
- Access to AWS Management Console
 - Basic understanding of databases
 - Basic familiarity with MySQL commands



Part 1: Create an RDS Database Instance

1. Open the **AWS Management Console** and switch to the us-east-1 region
2. Navigate to the **RDS service**
3. Click "**Create database**"
4. Choose the following settings:
 - Choose "**Standard create**"
 - Engine type: **MySQL**
 - Version: **MySQL 8.0.28** (or latest available version)
 - Template: **Free tier**



- DB instance identifier: **mylab-database**
- Master username: **admin**
- Master password: **Create a secure password** and note it down
- DB instance class: Select

Burstable classes (includes t3) ○ Choose **db.t3.micro** ○

Storage: **20 GB (GP2)** ○ Uncheck "**Enable storage**

autoscaling" ○ Under "**Connectivity**":

✦ **VPC**: Choose default VPC

✦ **Public access**: Yes

✦ **VPC security group**: Create new

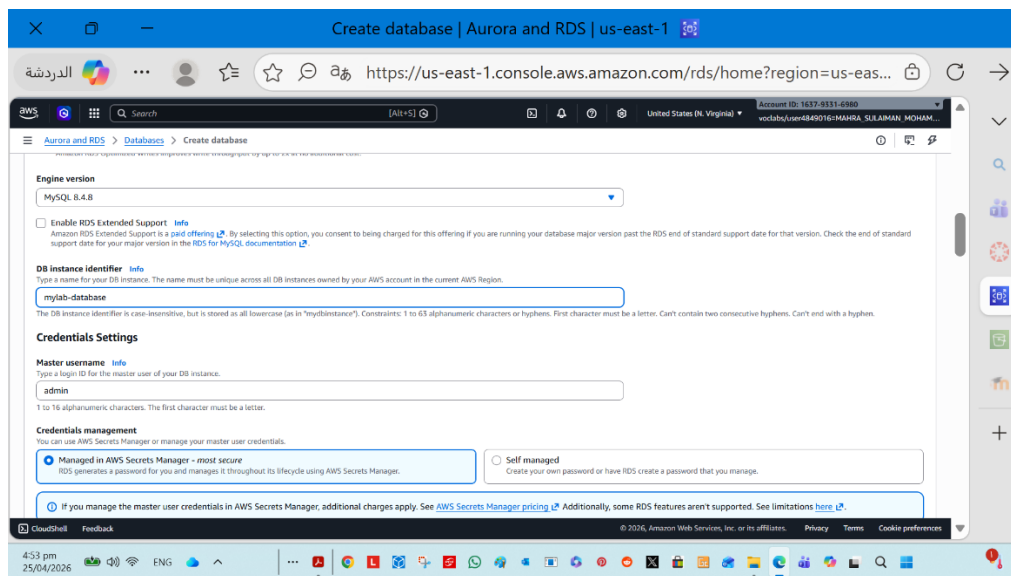
✦ **New security group name**: rds-lab-sg ○ Under "**Additional configuration**":

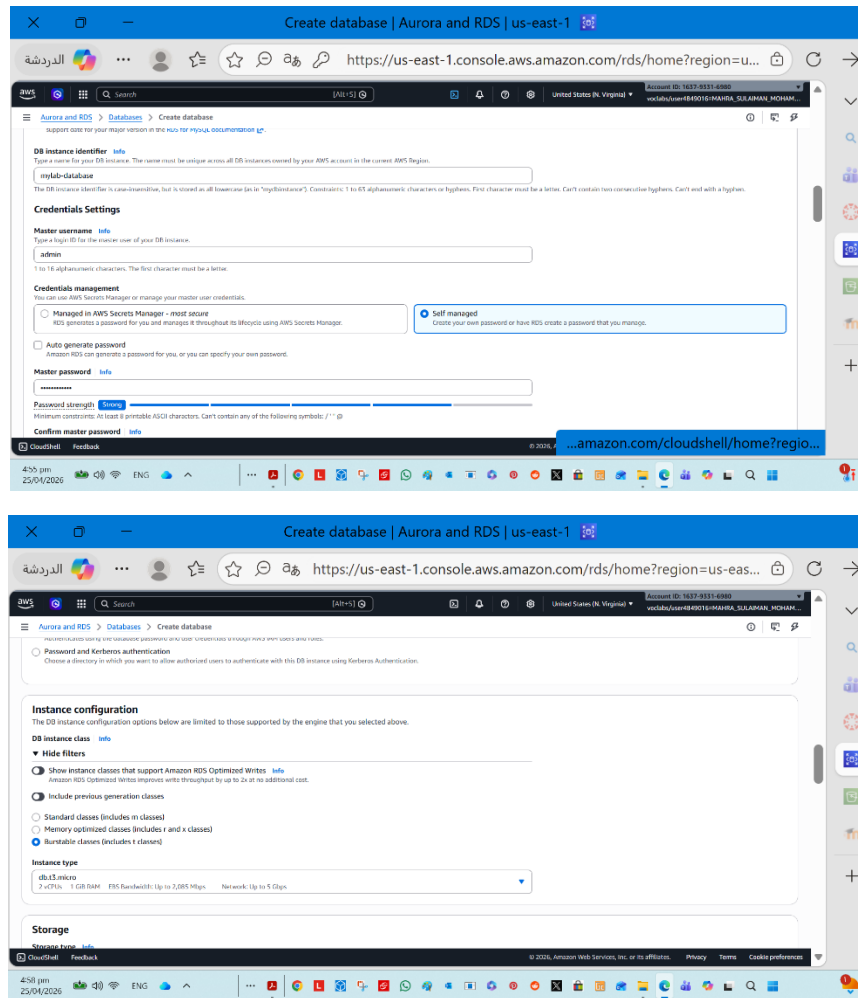
✦ Initial database name: **labdb**

✦ **Uncheck** "Enable automated backups"

✦ **Uncheck** "Enable Enhanced monitoring"

5. Click "**Create database**"

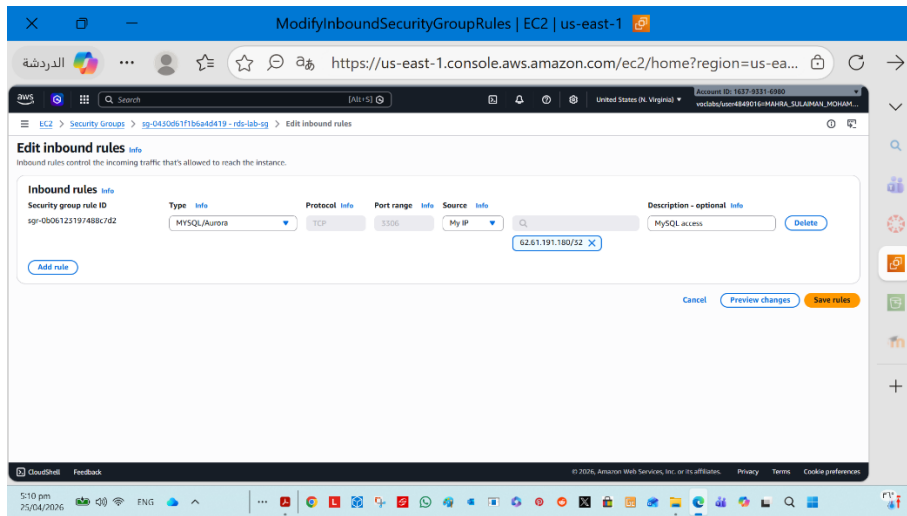




Part 2: Configure Security Group

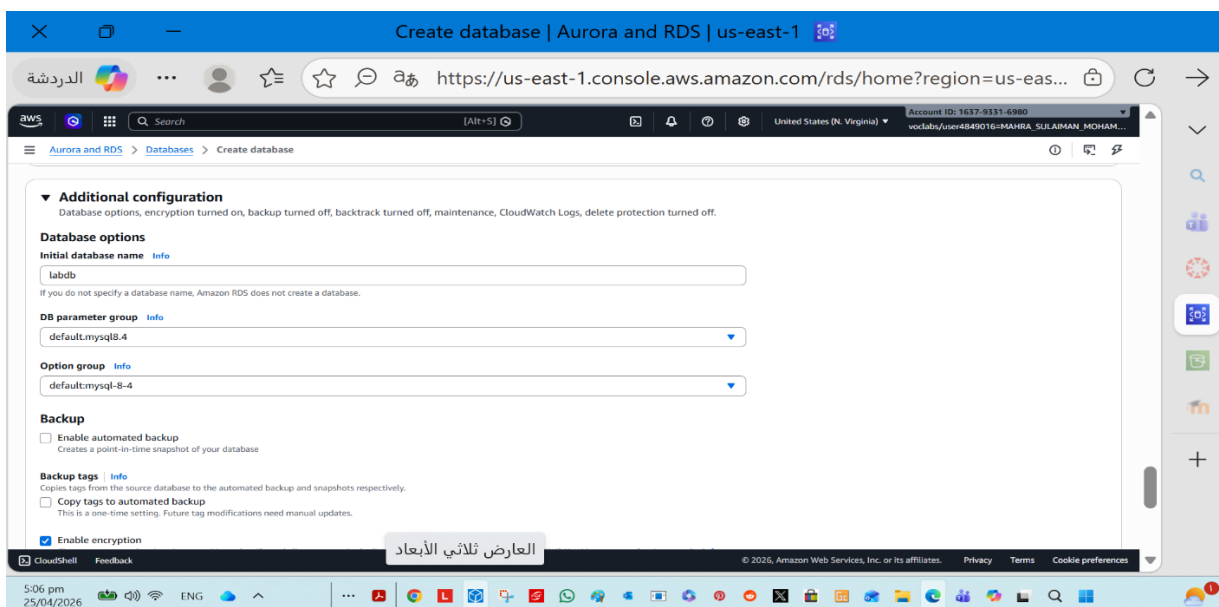
1. While the database is creating (takes ~5-10 minutes), navigate to EC2 > Security Groups
2. Find the security group created for your RDS instance (**rds-lab-sg**)
3. **Add an inbound rule:**
 - **Type:** MySQL/Aurora (3306) ○
 - Source:** Your IP address (select "My IP") ○
 - Description:** MySQL access
4. Click "**Save rules**"

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Part 3: Connect to Database

1. Once the RDS instance status shows "**Available**", click on the instance ID
2. Note down the following information:
 - Endpoint
 - Port (3306)
 - Master username (admin)
 - Database name (labdb)



To connect from MySQL Workbench

1. Download and install MySQL Workbench at [Download MySQL Workbench](#) Open MySQL Workbench.



- From **Database**, choose **Manage Connections**.
- In the **Manage Server Connections** window, choose **New**.
- In the **Connect to Database** window, enter the following information:
- **Stored Connection** – Enter a name for the connection, such as MyDB.
- **Hostname** – Enter the DB instance endpoint.
- **Port** – Enter the port used by the DB instance.
- **Username** – Enter the user name of a valid database user, such as the master user.
- **Password** – Optionally, choose **Store in Vault** and then enter and save the password for the user.

The window looks similar to the following:

Manage Server Connections

MySQL Connections

- awsmysql
- dmsoracle
- testconnect
- MyDB**
- new connection
- MyDB

Connection Name:

Connection Remote Management System Profile

Connection Method: Method to use to connect to the RDBMS

Parameters SSL Advanced

Hostname: Port: Name or IP address of the server host - and TCP/IP port.

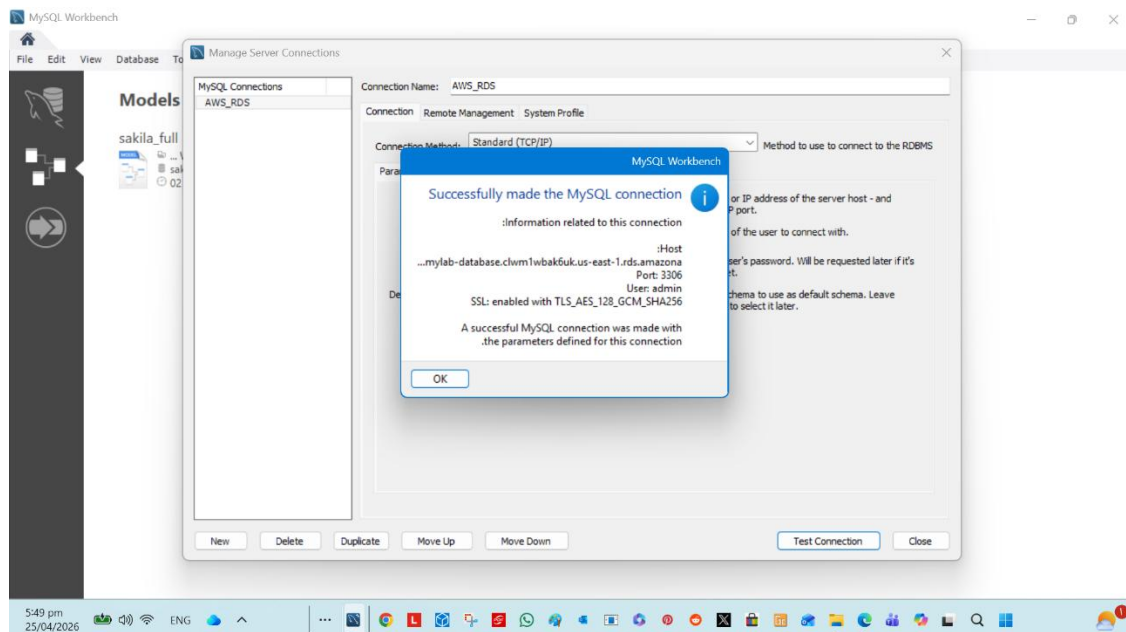
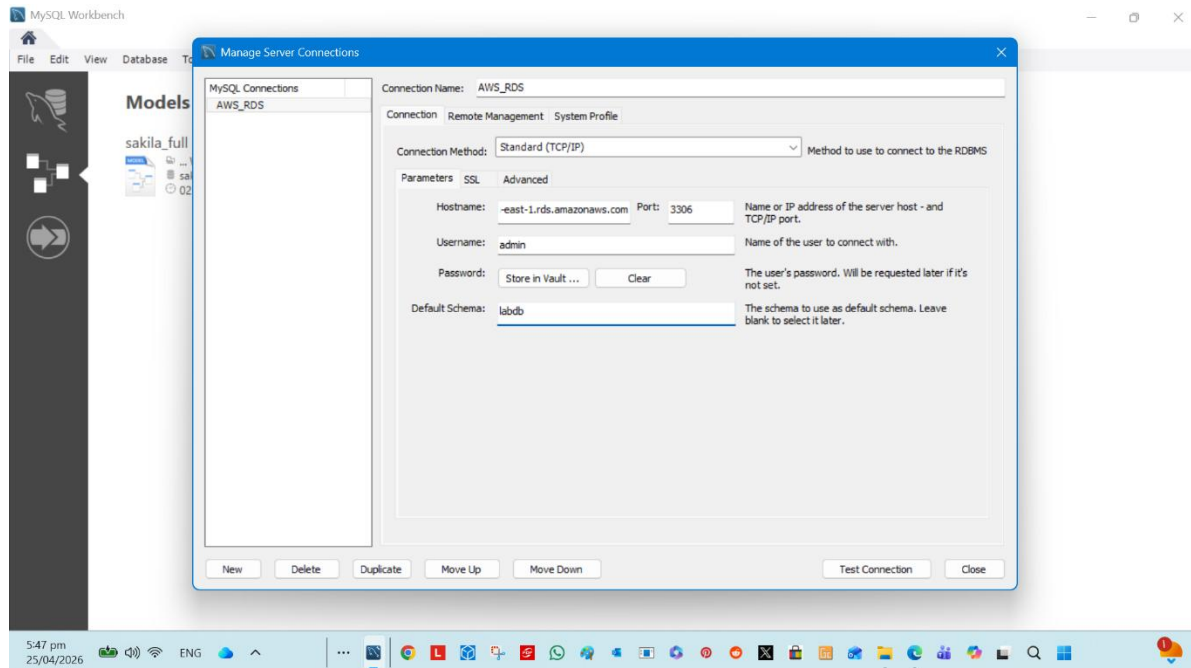
Username: Name of the user to connect with.

Password: The user's password. Will be requested later if it's not set.

Default Schema: The schema to use as default schema. Leave blank to select it later.

- Optionally, choose **Test Connection** to confirm that the connection to the DB instance is successful. ○ Choose **Close**.
- From **Database**, choose **Connect to Database**. ○ From **Stored Connection**, choose your connection. ○ Choose **OK**.

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Part 4: Basic Database Operations

```
CREATE DATABASE labdb;
```

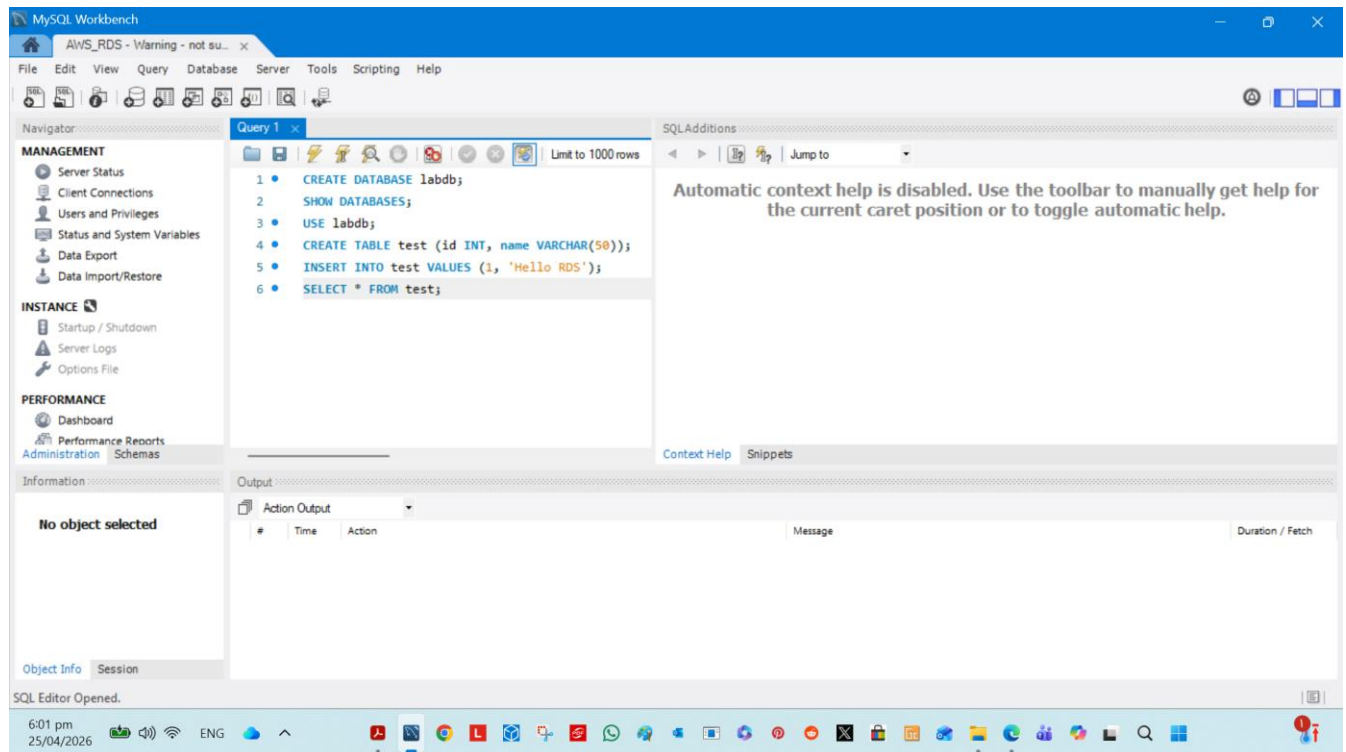
```
SHOW DATABASES;
```

```
USE labdb;
```

```
CREATE TABLE test (id INT, name VARCHAR(50));
```

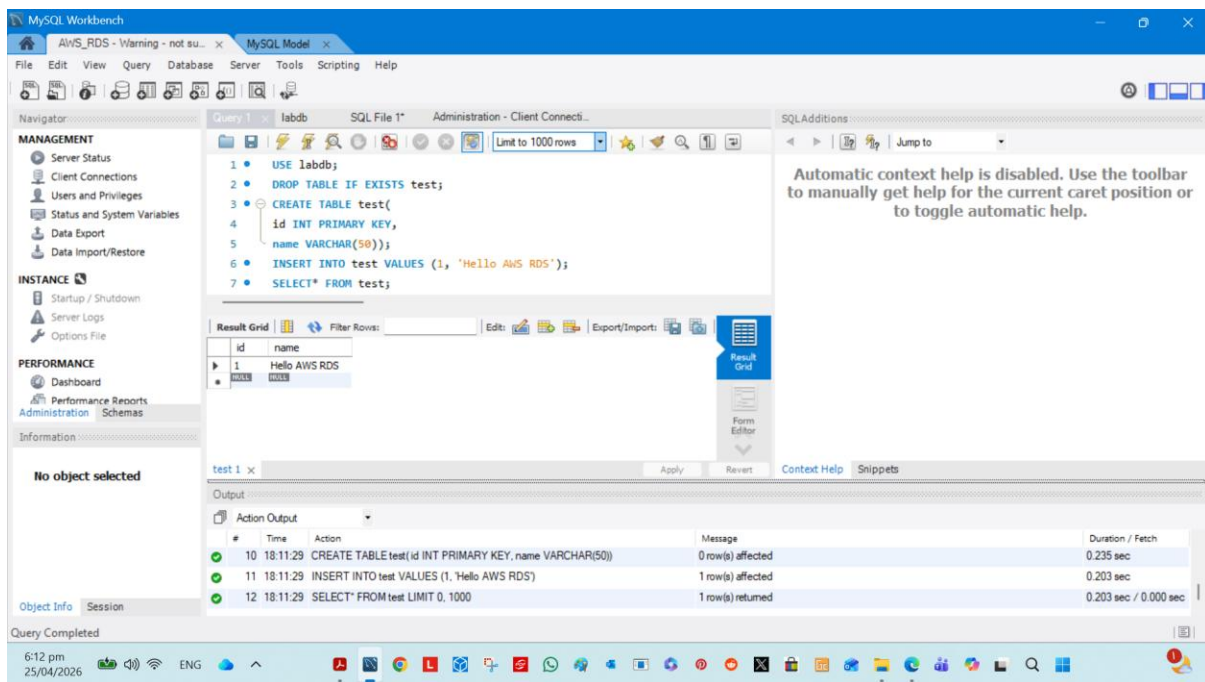
```
INSERT INTO test VALUES (1, 'Hello RDS');
```

```
SELECT * FROM test;
```

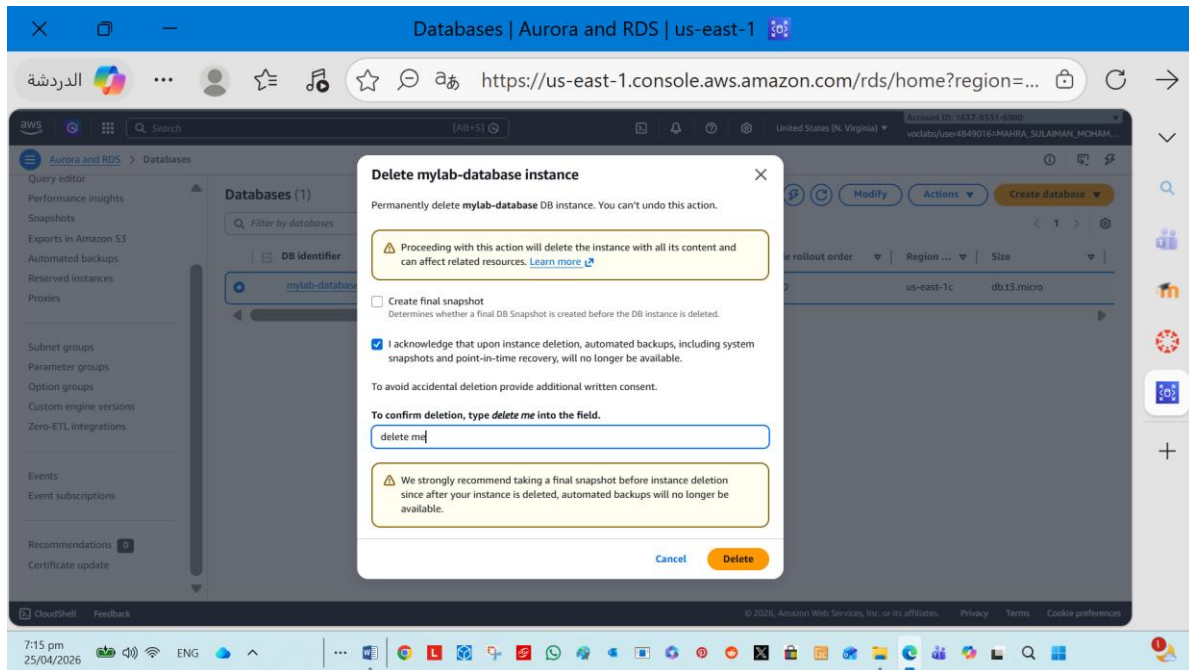


CLEAN-UP STEPS

1. Return to RDS Console
2. Select your database instance
3. Click "Actions"
4. Select "Delete"
5. Uncheck "Create final snapshot"
6. Uncheck "Retain automated backups"
7. Select "I Acknowledge"
8. Type "delete me" in the confirmation box
9. Click "Delete"



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